

PUPIL: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 6 ENQUIRY HOURS • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Evidence: *Stretcher Bearers of the RAMC Lifting a Wounded Man out of a Trench* • Gilbert Rogers (c. 1919)

OFFICIAL EDEXCEL SOURCE ARCHIVE • IWM ART 2485 / RAMC HISTORIC COLLECTION

1. FEATURE DRILLS [4M]

Q1(a) & Q1(b): Two 2-mark questions. Identify 1 specific feature + 1 supporting factual detail (5 mins).

2. SOURCE UTILITY [8M]

Q2(a): 3-step evaluation of Content, Provenance (Nature, Origin, Motive), and Contextual Limitations (10 mins).

3. FOLLOW-UP ENQUIRY [4M]

Q2(b): 4-row tabular enquiry: Detail, Historical Question, Source Type, and Clinical Value (5 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918)	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles <i>Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?</i>	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses <i>How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?</i>	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection <i>Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?</i>	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS <i>How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?</i>	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays <i>How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?</i>	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction <i>How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?</i>	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT 1914 • Flanders Mud & The 1st Battle of Ypres

Key Topic 5.1

The British Expeditionary Force (BEF) holds the Ypres Salient. Low-lying Flanders clay creates a dangerously high water table; trenches fill with liquid mud. Shelling smashes all natural drainage canals, leaving men standing waist-deep in water and giving rise to the first epidemics of trench foot and dysentery.

APR 1915 • 2nd Battle of Ypres: The First Poison Gas Attack

Key Topic 5.2 • 5.3

German forces release 160 tons of chlorine gas across a 4-mile front. British troops suffer suffocation and fluid in the lungs; soldiers improvise with urine-soaked pads before chemically impregnated cotton-pad respirators (the Hypo Helmet) are rushed to the front. Lice-borne trench fever breaks out across all divisions.

DEC 1915 • The Thomas Splint & Introduction of the Brodie Helmet

Key Topic 5.3 • 5.5

Robert Jones introduces the Thomas Splint to frontline medical posts, stabilizing compound fractures and reducing femur mortality from 80% to 20%. Simultaneously, the War Office begins issuing John Brodie’s stamped steel helmet, reducing penetrating shrapnel head trauma by an estimated 75–80%.

LIVING TIMELINE • PART 2: MASS OFFENSIVES & CLINICAL INNOVATION (1916–1917)

Instructions: Complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • The Battle of the Somme: The Triage Crisis

Key Topic 5.4

The BEF suffers 57,000 casualties on 1 July 1916. Stretcher bearers and Casualty Clearing Stations are overwhelmed; motor ambulances struggle along single rutted roads. As a direct result, the RAMC enforces a strict three-tier triage system (Walking Wounded, Immediate Surgery, Moribund) to prevent total collapse.

APR–NOV 1917 • Arras Underground Hospital & Passchendaele Mud

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

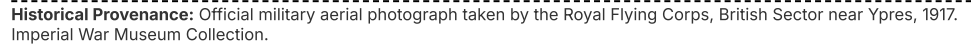
At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?

Specification Requirement: The British sector of the Western Front: theatre of war in Flanders and northern France; Ypres, Somme, Arras, Cambrai; trench system (frontline, support, reserve, communication trenches; saps); terrain, clay, water table, mud; underground systems at Arras (Thompson's Cave).

1 1914–1915 Flanders Terrain & Ypres Salient • Low-lying Flanders clay: very high water table • Heavy rain and artillery blast destroy drainage • British held vulnerable salient overlooked by Germans • Liquid mud made stretcher evacuation agonizingly slow FOCUS CLUE Why was the low-lying topography of Ypres a tactical and medical nightmare?	
2 1914–1918 Trench Architecture & Traverses • Trenches dug 2m deep in zig-zag traverse pattern • Traverses contain shell blast and flying shrapnel • Prevents enemy firing straight down length of trench • Duckboards, fire-steps, parapets, and parados fitted FOCUS CLUE How did zig-zag traverses limit shrapnel injuries and contain shell explosions?	
3 1915–1918 Trench System: Front to Rear • Frontline: fire-steps for observation and combat • Support trench (70m back) & Reserve (hundreds m) • Communication trenches move fresh men & stretchers • Saps: narrow blind tunnels into No Man's Land FOCUS CLUE Why did narrow communication trenches create acute bottlenecks for stretchers?	
4 Nov 1916–1917 Thompson's Cave & Arras Tunnels • Chalk geology enabled massive tunnel excavations • New Zealand tunnelers joined ancient Roman quarries • Electric lighting, running piped water & rail tracks • 700-bed underground hospital (Thompson's Cave) FOCUS CLUE How did the chalk geology at Arras allow safe, sterile surgical conditions?	
5 1916–1917 Mass Trauma: Somme & Cambrai • July 1916 Somme: 57,000 British casualties Day 1 • Stretcher bearers overwhelmed by mass casualties • Nov 1917 Cambrai: mass tank assault over chalk ridge • Mobile breakthrough demanded rapid surgical shifts FOCUS CLUE Why did mass-casualty offensives force the RAMC to reorganize evacuation?	

ORIGIN: 1917



2 SEPARATE QUESTIONS • 2 MARKS EACH

Hint: Consider the chalk geology, lighting, water supply, and 700-bed surgical capacity 20 metres underground.

A03 EVALUATION

Royal Flying Corps • zig-zag traverses • communication trench • Ypres Salient
• Messines Ridge • water table • duckboards • parapet • parados • fire-step

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

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Write your formal answers on Pages 6–7 →

OFFICIAL EXAMINATION RESPONSE • QUESTION 1 & QUESTION 2

Q1 Total: [/ 4]

[/2]

DETAIL: _____

[/2]

DETAIL: _____

Continue on next page →

[Begin Question 2(a) Utility Evaluation • Content • Provenance • Context]

[illegible]

KT5.1: The Theatre of War: The British Sector, Trench Geography & Battles

SUSTAINED EVALUATION • DISCIPLINARY CONTEXT • HISTORIC ENVIRONMENT

**SUSTAINED EVALUATION • DISCIPLINARY CONTEXT •
HISTORIC ENVIRONMENT**

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

[illegible]

Q2 Score: [/ 8] **Lesson Total:** [/ 12] **Teacher Feedback / Target:**

Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?

Specification Requirement: Ill health arising from the trench environment: trench foot (pathology, gangrene, prevention, whale oil, sock inspection); trench fever (body lice, *Pediculus humanus*, delousing stations, Serbian barrels); shell shock (symptoms, NYDN, rest treatment); dysentery (water chlorination, chloride of lime, latrines); underground shelter conditions.

1 1914–1915

Trench Foot Pathology

- Caused by standing in cold water and mud in tight boots
- Constricts capillary blood flow; tissues lose oxygen
- Numbness, severe swelling, skin turning blue and black
- Led to wet gangrene, tissue necrosis and amputation

FOCUS CLUE

How did prolonged cold water immersion destroy tissue and produce gangrene?

2 1915–1918

Prevention: Whale Oil & Socks

- Mandatory buddy system: soldiers rubbed whale oil daily
- Whale oil formed water-repellent barrier on skin
- Daily foot inspections strictly enforced by officers
- Every soldier carried three pairs of clean, dry socks

FOCUS CLUE

Why did mandatory officer inspections succeed where voluntary advice had failed?

3 1914–1918

Trench Fever, Lice & Delousing

- High fever, violent shivering, severe joint and shin pain
- Incapacitated up to 15% of troops for a month or more
- 1918: Proven transmitted by body lice (*Pediculus humanus*) faeces
- Divisional delousing bathhouses, steam ovens & Serbian barrels

FOCUS CLUE

Why did trench fever persist until the body louse insect vector was proven in 1918?

4 1914–1918

Dysentery & Water Chlorination

- Severe bacterial diarrhoea caused by unhygienic water & food
- Deep latrine pits dug away from frontline trenches
- Chloride of lime added daily to neutralize foul waste
- Drinking water purified with chemical chlorination tablets

FOCUS CLUE

How did strict sanitation and chemical water chlorination prevent epidemics?

5 1914–1918

Shell Shock & Psychological Trauma

- Caused by artillery blast concussion and mortal combat terror
- Symptoms: uncontrollable shaking, hysterical blindness, mutism
- ~80,000 British cases diagnosed; labeled NYDN (Not Yet Diagnosed, Nervous)
- Treated with rest and food; specialist centres like Craiglockhart

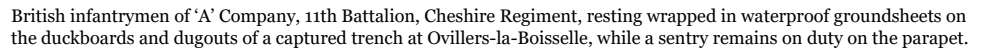
FOCUS CLUE

Why was shell shock initially stigmatised as cowardice before being treated medically?

ges 2–3 (Milestone 2). Draw soldiers applying whale oil and delousing garments in steam ovens.

← Pages 2–3

ORIGIN: July 1916



2 SEPARATE QUESTIONS • 2 MARKS EACH

Hint: Identify the parasite vector (body lice) and how lice faeces entered scratched skin to transmit the pathogen.

AO3 HISTORICAL ENQUIRY

Official Edexcel Prompt	High-Scoring Model Strategy • Historical Exemplar
1. Detail in Source A to follow up:	<i>"Soldiers huddled asleep on the damp trench floor wrapped in waterproof groundsheets."</i>
2. Question I would ask:	<i>What proportion of frontline infantry casualties were evacuated due to non-combat illnesses like trench fever and exhaustion rather than enemy action?</i>
3. Type of source I would use:	<i>RAMC Divisional Medical Officer monthly sickness returns and battalion casualty logs for the Somme sector in 1916.</i>
4. How this would help me:	<i>This would provide official statistical data proving how many soldiers were incapacitated by trench living conditions, vermin, and exhaustion compared to battle wounds.</i>

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

[illegible]

Write your formal answers on Pages 10–11 →

KT5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses

OFFICIAL EXAMINATION RESPONSE • QUESTION 1 & QUESTION 2

• **QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS]**

Q1 Total: [/ 4]

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front.

[12]

FEATURE:

DETAIL:

1(b) Describe one feature of the causes of trench fever among British troops.

[/ 2]

FEATURE:

DETAIL:

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS]

Score: [/ 4]

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry]

KT5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses

**SUSTAINED EVALUATION • DISCIPLINARY CONTEXT •
HISTORIC ENVIRONMENT**

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

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Q2 Score: [/ 4] **Lesson Total:** [/ 8] **Teacher Feedback / Target:**

How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?

Specification Requirement: Wounds, injuries and diseases: high-explosive artillery shells, shrapnel fragments; infection (gas gangrene, *Clostridium welchii*, tetanus); head trauma and the Brodie helmet; chemical gas attacks (chlorine, phosgene, mustard gas) and respirators.

1 1914–1918

Artillery & Shrapnel Trauma

- High-explosive artillery caused ~58% of all battle wounds
- Shells shattered into thousands of jagged steel splinters
- Smashed complex bones, caused blast shock and severe bleeding
- Mud, dirt, and filthy uniform cloth driven deep into tissue

FOCUS CLUE

Why did jagged shrapnel fragments cause far worse infections than rifle bullets?

2 1914–1918

Soil Bacteria & Gas Gangrene

- Manured Flanders farmland heavily contaminated with microbes
- *Clostridium welchii* bacterium produces lethal toxins and gas
- Anaerobic: flourishes in deep, airless shrapnel wounds
- Gas bubbles under skin; gangrene killed soldiers in hours

FOCUS CLUE

Why was *Clostridium welchii* so lethal in deep, airless shrapnel wounds?

3 Late 1915

Head Trauma & Brodie Helmet

- Soft cloth caps offered zero protection against shrapnel
- Late 1915: John Brodie patented the steel combat helmet
- Stamped from single steel sheet; wide rim deflected blast
- Reduced fatal penetrating shrapnel head wounds by ~80%

FOCUS CLUE

Why did the introduction of the Brodie helmet cause an apparent rise in hospital head cases?

4 Apr 1915–1916

Chlorine & Phosgene Attacks

- Apr 1915: Germans used chlorine gas at 2nd Battle of Ypres
- Choking gas destroyed respiratory lungs; fluid suffocation
- Urine pads replaced by chemical hypo hoods and P-helmets
- Dec 1915: Phosgene introduced (faint musty smell; 6x deadlier)

FOCUS CLUE

How did the rapid evolution of chemical respirators counter suffocating gases?

5 1916–1918

Mustard Gas & Box Respirators

- 1916: Small Box Respirator issued (carbon/charcoal filter)
- July 1917: Germans deployed mustard gas at 3rd Ypres
- Blistering agent burned skin through uniforms; caused blindness
- Odorless, oily liquid soaked into soil; persisted for weeks

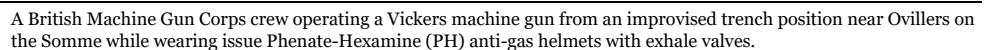
FOCUS CLUE

Why was mustard gas far harder to counter than chlorine or phosgene?

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Sketch the Brodie steel helmet and British PH gas hood.

← Pages 2–3

ORIGIN: July 1916



Historical Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke near Ovillers during the Battle of the Somme, July 1916. Imperial War Museum Collection.

2 SEPARATE QUESTIONS • 2 MARKS EACH

Hint: Consider the pressed-steel construction, shallow brim, and internal lining to deflect shrapnel from above.

A03 EVALUATION

high-explosive artillery • shrapnel balls • gas gangrene • *Clostridium welchii* • Brodie steel helmet • chlorine gas (1915) • phosgene • mustard gas (1917) • PH helmet • Small Box Respirator (1916)

Write your formal answers on Pages 14–15 →

OFFICIAL EXAMINATION RESPONSE • QUESTION 1 & QUESTION 2

Q1 Total: [/ 4]

[12]

[/2]

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KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection SUSTAINED EVALUATION • DISCIPLINARY CONTEXT • HISTORIC ENVIRONMENT

SUSTAINED EVALUATION • DISCIPLINARY CONTEXT • HISTORIC ENVIRONMENT

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

[illegible]

Q2 Score: [/ 8] Lesson Total: [/ 12] Teacher Feedback / Target:

How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?

Specification Requirement: The work of the RAMC and FANY: the Chain of Evacuation (stretcher bearers, Regimental Aid Posts, Field Ambulances and Dressing Stations, Casualty Clearing Stations, Base Hospitals); triage systems; transport methods (motor ambulances, ambulance trains, canal barges).

1 1914–1918

Stretcher Bearers & RAP

- 16 bearers per battalion (4 per stretcher; 6–8 in mud)
- Recovered wounded under direct machine-gun/artillery fire
- Regimental Aid Post (RAP) 200m behind front line in dugout
- Regimental Medical Officer (RMO) applied iodine & bandages

FOCUS CLUE

Why was rapid evacuation by stretcher bearers vital to prevent irreversible shock?

2 1914–1918

Field Ambulances & ADS/MDS

- Staffed by Royal Army Medical Corps (RAMC) Field Ambulance
- Advanced Dressing Station (1/2 mile back); MDS (1 mile back)
- Located in abandoned farmhouses, church ruins, or bunkers
- Minor wounds dressed; severe cases dispatched by ambulance

FOCUS CLUE

What was the tactical role of Dressing Stations in preventing CCS congestion?

3 1914–1918

CCS & The Triage System

- Located 7–12 miles behind front, near railway junctions
- Primary surgical centre for critical life-saving operations
- Triage: 1. Walking Wounded, 2. Immediate, 3. Moribund
- Major amputations and emergency abdominal laparotomies done

FOCUS CLUE

How did the 3-tier triage system maximize survival during mass offensives?

4 1914–1918

Base Hospitals on French Coast

- Located at coastal ports (Boulogne, Calais, Le Havre, Étaples)
- Large civilian hospitals, hotels, or vast tent cities
- Specialized wards: amputees, gas burns, chest injuries
- Patients convalesced or shipped back to Britain on hospital ships

FOCUS CLUE

Why were Base Hospitals situated near major coastal rail terminals and ports?

5 1914–1918

Motor Ambulances & Barges

- Horse ambulances caused agony; shaken by rough shell-holes
- 1914 Red Cross appeal raised 512 motorized ambulances
- Hospital trains fitted with kitchens and operating rooms
- Canal barges provided smooth travel for head and chest trauma

FOCUS CLUE

Why were canal barges preferred over trains for head and chest trauma?

TIMELINE MISSION Turn to Pages 2–3 (Milestone 5). Draw the 6 sequential stages of the Chain of Evacuation.

← Pages 2–3

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS**SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT****ORIGIN:** August 1917

A team of six British stretcher bearers struggling through deep, liquid Flanders mud to carry a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

Historical Provenance: Official British Army photograph taken by Lieutenant John Warwick Brooke near Boesinghe during the Third Battle of Ypres, 1 August 1917. Imperial War Museum Collection.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]**2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. [2 marks]

Hint: Explain how casualties were sorted into Walking Wounded, Immediate Surgery, and Moribund to maximize survival.

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. [2 marks]

Hint: Consider ambulance driving, mobile soup kitchens, and transporting wounded through artillery fire.

• QUESTION 2(B): 4-PROMPT FOLLOW-UP ENQUIRY [4 MARKS • 5 MINS]**AO3 HISTORICAL ENQUIRY**

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

Official Edexcel Prompt	High-Scoring Model Strategy • Historical Exemplar
1. Detail in Source A to follow up:	"A team of six stretcher bearers struggling through deep, liquid mud to evacuate a wounded soldier on a wooden stretcher near Boesinghe."
2. Question I would ask:	What was the average time taken to evacuate a casualty from the frontline sap to the RAP and Dressing Station during the Third Battle of Ypres?
3. Type of source I would use:	RAMC Battalion Medical Officer war diaries and Field Ambulance stretcher bearer logs from the Ypres sector in autumn 1917.
4. How this would help me:	This would reveal exact transport durations, bearer casualty rates, and whether delays caused fatal haemorrhagic shock.

Key Vocabulary: RAMC • FANY • Regimental Aid Post • Advanced Dressing Station • Casualty Clearing Station • Base Hospital • triage system • motor ambulance • hospital train • canal barge

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:**FORMAL EXAMINATION PRACTICE****Write your formal answers on Pages 18–19 →**

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS

OFFICIAL EXAMINATION RESPONSE • QUESTION 1
& QUESTION 2

• **QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS]**

Q1 Total: [/ 4]

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front.

[12]

FEATURE:

DETAIL:

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front.

[12]

FEATURE:

DETAIL:

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS]

Score: [/ 4]

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[*Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry*]

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS

**SUSTAINED EVALUATION • DISCIPLINARY CONTEXT •
HISTORIC ENVIRONMENT**

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

[illegible]

Q2 Score: [/ 4] **Lesson Total:** [/ 8] **Teacher Feedback / Target:**

Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?

Specification Requirement: Medical advances on the Western Front: the Thomas Splint (Hugh Owen Thomas, Robert Jones) reducing compound femur mortality; wound debridement and delayed primary closure; the Carrel-Dakin antiseptic irrigation method; mobile X-ray units and radiology.

1 1914–1918

Wound Debridement & Excision

- Surface antiseptics failed against deep dirty shrapnel wounds
- Surgeons cut away all dead, bruised, and infected tissue
- Deprived anaerobic bacteria (*Clostridium welchii*) of dead flesh
- Delayed primary closure: wound left open to flush before stitching

FOCUS CLUE

Why was surgical excision of dead tissue more effective than chemical?

2 1915–1918

Carrel-Dakin Antiseptic Flush

- Alexis Carrel & Henry Dakin design continuous wound irrigation
- Dilute sodium hypochlorite (buffered chlorine solution)
- Flushed into deep wound through network of perforated rubber tubes
- Prevented gangrene; degraded quickly so made fresh every 24h

FOCUS CLUE

How did continuous chemical irrigation solve the problem of deep wound infection?

3 Dec 1915–1916

The Thomas Splint Breakthrough

- Compound femur fracture mortality in 1914 stood at 80%
- Muscle spasms pulled sharp broken bone ends through femoral artery
- Hugh Owen Thomas splint introduced to front by Robert Jones
- Secured leg in rigid mechanical traction; mortality dropped to 20%

FOCUS CLUE

How did rigid traction prevent fatal haemorrhage and shock from broken femurs?

4 1914–1918

Mobile X-Rays & Marie Curie

- Essential to locate shrapnel and bullets before surgical incision
- Marie Curie equipped and deployed 20 mobile X-ray vans
- Powered by van engines; operated near Casualty Clearing Stations
- Static base hospital X-rays located deep fragments and gas pockets

FOCUS CLUE

Why was locating shrapnel on an X-ray vital before operating on gas gangrene?

5 1914–1918

X-Ray Limits in Combat

- Glass X-ray tubes fragile; easily shattered on rutted roads
- Tubes overheated after minutes; required cooling periods
- Could not detect cloth, mud, or wooden splinters in tissue
- Severely wounded soldiers had to lie completely still for minutes

FOCUS CLUE

Why did undetected cloth fragments remain the primary cause of post-op gangrene?

TIMELINE MISSION Turn to Pages 2–3 (Milestone 3). Sketch the Thomas Splint groin ring and traction cord.

← Pages 2–3

KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays**SOURCE A • CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE**

ORIGIN: c.1916



An authentic surviving British Army Thomas Splint, showing the padded oval ring fitted against the groin, rigid metal side bars, and canvas support straps used to maintain traction and immobilize broken thigh bones.

Historical Provenance: Original First World War Thomas Splint preserved in the Wellcome Historical Medical Collection, London. Designed by Hugh Owen Thomas and introduced to the Western Front in 1915–1916 by Robert Jones.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]**2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the Thomas Splint used on the Western Front. [2 marks]

Hint: Explain the padded metal ring fitted against the groin and the windlass traction cord pulling the leg taut.

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. [2 marks]

Hint: Explain the system of perforated rubber tubes irrigating the deep wound with sodium hypochlorite antiseptic.

• QUESTION 2(A): SOURCE UTILITY ASSESSMENT [8 MARKS • 10 MINS]**AO3 EVALUATION**

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into limb treatments because... • Specifically, the photograph demonstrates... • In terms of provenance, as an RAMC instructional image... • However, the source is limited because... • Therefore...

KEY VOCABULARY:

Thomas Splint • Hugh Owen Thomas • Robert Jones • compound fracture • femoral artery • mechanical traction • wound debridement • Carrel-Dakin solution • delayed primary closure • mobile X-ray unit

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:**FORMAL EXAMINATION PRACTICE**

Write your formal answers on Pages 22–23 →

Q1 Total: [/ 4]

[12]

[/2]

Continue on next page →

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KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays

**SUSTAINED EVALUATION • DISCIPLINARY CONTEXT •
HISTORIC ENVIRONMENT**

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

[illegible]

Q2 Score: [/ 8] Lesson Total: [/ 12] Teacher Feedback / Target:

How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?

Specification Requirement: Medical advances on the Western Front: blood transfusions and storage (Landsteiner, Hustin, Rous and Turner, Captain Oswald Robertson and the Cambrai blood bank); specialized neurosurgery (Harvey Cushing); plastic and facial reconstruction (Harold Gillies, Queen's Hospital Sidcup, tubed pedicle).

1 1914–1915

Sodium Citrate & Clotting

- 1914: Albert Hustin discovers sodium citrate stops clotting
- Prevents blood from coagulating when exposed to air
- 1915: Richard Lewisohn determines safe non-toxic dosage (0.2%)
- Enabled indirect transfusion: donor no longer strapped to patient

FOCUS CLUE

How did sodium citrate liberate blood transfusion from donor-to-patient tubing?

2 1916–1917

Citrate–Glucose Blood Storage

- Francis Rous & James Turner add glucose (dextrose) solution
- Preserves fragile red blood cells from breaking down
- Citrated blood kept in ice chests remained viable up to 4 weeks
- Made storing blood possible in advance of military offensives

FOCUS CLUE

Why was glucose addition essential to prevent red blood cell breakdown?

3 Nov 1917

Cambrai Blood Bank Depot

- Capt Oswald Hope Robertson created world's first blood bank
- Collected universal Group O blood from healthy walking wounded
- Stored in 22 iced metal vacuum flasks at Casualty Clearing Station
- Treated 20 severely shocked soldiers at Cambrai; 11 survived

FOCUS CLUE

Why did pre-stored blood transform surgical survival during surprise assaults?

4 1917–1918

Harvey Cushing's Brain Surgery

- American surgeon Harvey Cushing pioneered delicate military neurosurgery
- Used local anaesthetic instead of general to prevent fatal brain swelling
- Silver clips clamped vessels (haemostasis); magnets extracted deep shrapnel
- Operated on 45 patients at 3rd Ypres, cutting mortality from 54% to 29%

FOCUS CLUE

Why was local anaesthesia superior to general anaesthetic in head trauma?

5 1917–1918

Harold Gillies at Sidcup

- Harold Gillies founded Queen's Hospital, Sidcup, Kent (1917)
- Treated over 11,000 men with shattered faces and jaw trauma
- Pioneered tubed pedicle skin grafting to maintain active blood supply
- Grafts from chest/neck rebuilt noses and jaws; foundation of plastic surgery

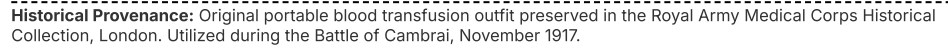
FOCUS CLUE

How did the tubed pedicle technique prevent skin grafts from rotting and dying?

TIMELINE MISSION Turn to Pages 2–3 (Milestone 6). Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle.

← Pages 2–3

ORIGIN: 1917



• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS]

Q1 Total: [/ 4]

1(a) Describe one feature of Harvey Cushing’s surgical techniques in operating on head wounds on the Western Front.

[/ 2]

FEATURE:

DETAIL:

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen’s Hospital, Sidcup.

[/ 2]

FEATURE:

DETAIL:

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS]

Score: [/ 4]

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry]

KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction

**SUSTAINED EVALUATION • DISCIPLINARY CONTEXT •
HISTORIC ENVIRONMENT**

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

[illegible]

Q2 Score: [/ 4] **Lesson Total:** [/ 8] **Teacher Feedback / Target:**

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT5.1	Q1: Describe Two Features of Trenches / Arras [4m]	p. 6	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(a): Source Utility Assessment (Trench Design) [8m]	p. 6–7	/ 8	
KT5.2	Q1: Describe Two Features of Trench Foot / Fever [4m]	p. 10	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(b): 4-Prompt Follow-Up Matrix (Living Conditions) [4m]	p. 10–11	/ 4	
KT5.3	Q1: Describe Two Features of Shrapnel / Poison Gas [4m]	p. 14	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(a): Source Utility Assessment (Artillery Trauma) [8m]	p. 14–15	/ 8	
KT5.4	Q1: Describe Two Features of Dressing Station / FANY [4m]	p. 18	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(b): 4-Prompt Follow-Up Matrix (Stretcher Transport) [4m]	p. 18–19	/ 4	
KT5.5	Q1: Describe Two Features of Thomas Splint / X-Rays [4m]	p. 22	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(a): Source Utility Assessment (Mobile X-Rays) [8m]	p. 22–23	/ 8	
KT5.6	Q1: Describe Two Features of Cushing / Gillies Surgery [4m]	p. 26	/ 4	MARKING FEEDBACK • TARGETS: _____ _____ _____
	Q2(b): 4-Prompt Follow-Up Matrix (Cambrai Blood Bank) [4m]	p. 26–27	/ 4	

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10/20 Questions Per Enquiry

KT5.1 Trenches & Mud  LESSON HUB  MASTERY QUIZ Score: [/ 10]	KT5.2 Trench Illness  LESSON HUB  MASTERY QUIZ Score: [/ 10]	KT5.3 Trauma & Gas  LESSON HUB  MASTERY QUIZ Score: [/ 10]	KT5.4 Evacuation Chain  LESSON HUB  MASTERY QUIZ Score: [/ 10]	KT5.5 Splints & X-Rays  LESSON HUB  MASTERY QUIZ Score: [/ 10]	KT5.6 Blood & Plastic  LESSON HUB  MASTERY QUIZ Score: [/ 10]
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